PROJECT REPORT

DOOR STATUS INDICATOR

Schematic Design and PCB Implementation

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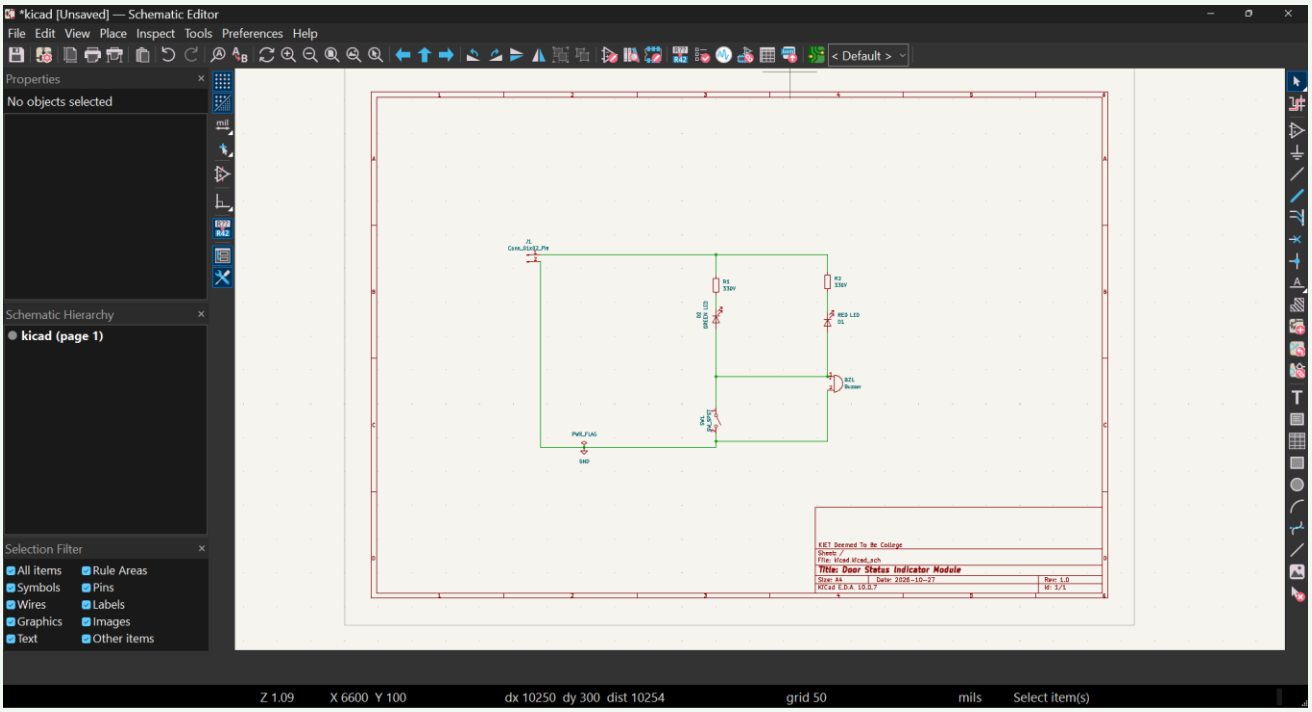
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Circuit Schematic

The schematic shows the electrical connections of the Door Status Indicator module. The design has a two-pin connector, two resistors, red and green LEDs, an SPST switch, and a buzzer. It serves as the electrical reference for the PCB layout.



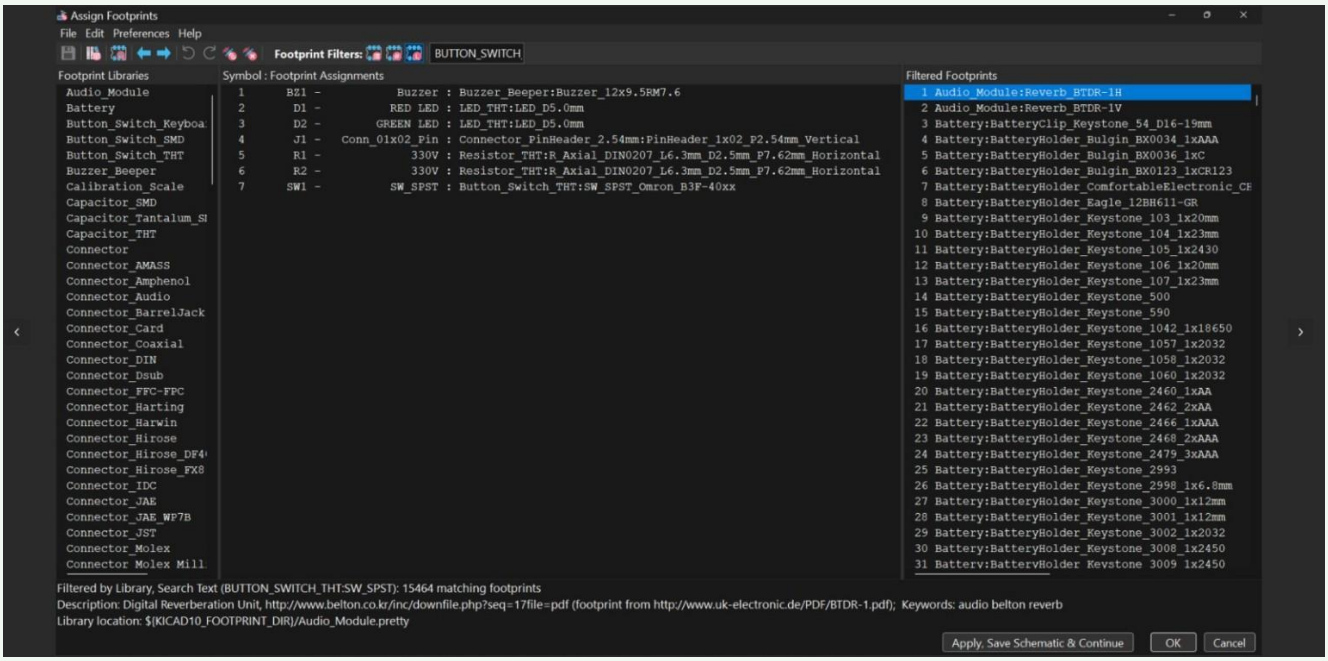
Circuit schematic captured in KiCad Schematic Editor.

Footprint Allocation

The footprint assignment step links each schematic symbol to the physical package used on the PCB. The screenshot below shows the assignments chosen in KiCad.

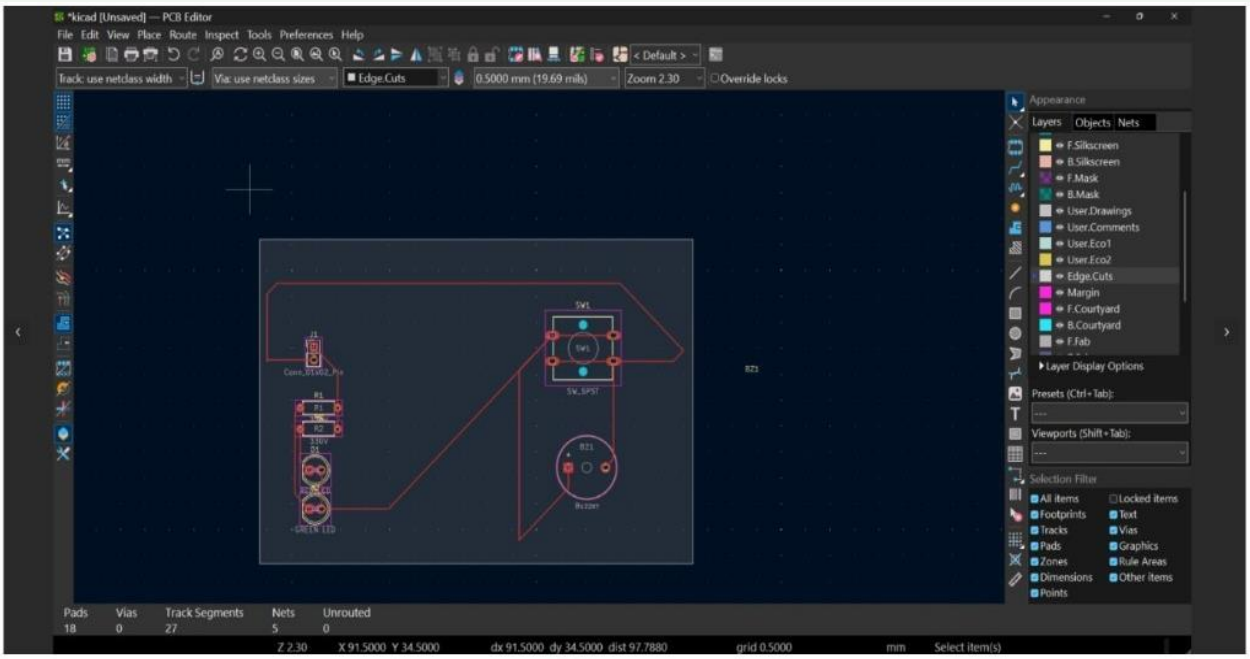
Ref.	Component	Assigned footprint
BZ1	Buzzer	Buzzer_Beeper:Buzzer_12x9.5RM7.6
D1	Red LED	LED_THT:LED_D5.0mm
D2	Green LED	LED_THT:LED_D5.0mm
J1	2-pin connector	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
R1, R2	Resistors	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
SW1	SPST switch	Button_Switch_THT:SW_SPST_Omron_B3F-40xx

Figure 2. KiCad Assign Footprints window.



2D PCB View — Routed Layout

The 2D PCB view displays the board outline, component placement, pads, and copper tracks. The supplied KiCad screenshot reports 18 pads, 27 track segments, 5 nets, and 0 unrouted connections in its status bar.



3D PCB Preview

The 3D preview gives a visual check of component placement, board shape, and orientation. The board shown contains the connector, two resistors, two LEDs, a switch, and a buzzer.

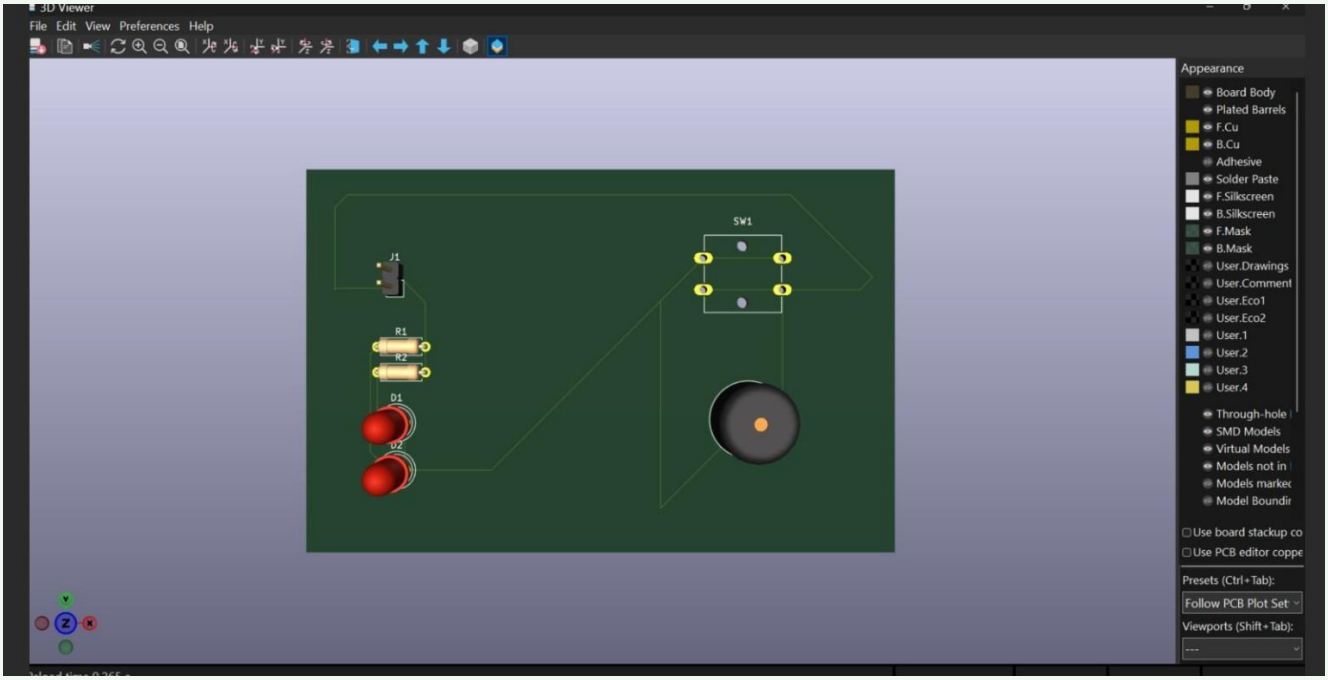


Figure 4. Three-dimensional PCB preview in KiCad 3D Viewer.

Operating Principle

The Door Status Indicator is meant to give a visible and audible signal of a door's state. The switch acts as the state input. Based on the switch position and the final circuit wiring, the red and green LEDs give a visual indication, while the buzzer sounds an audible alert. The resistors limit the current through the LEDs.

Before powering the circuit, verify the supply voltage, LED polarity, buzzer rating, switch connections, and resistor values. Confirm all electrical connections and run the PCB design-rule check before fabrication.

Conclusion

This project shows the PCB design workflow in KiCad, from schematic capture and footprint assignment through 2D routing to 3D visualization. The included screenshots record the key stages of the design. Final assembly should go ahead only after component compatibility, electrical values, and board clearances have been checked.